SOFTWARE REQUIREMENTS SPECIFICATION

AI-Driven Medical Second Opinion System

for Diagnostic Validation and Clinical Decision Support in Developing Healthcare Infrastructures

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1. INTRODUCTION

This Software Requirements Specification (SRS) establishes the complete functional and non-functional requirements for the AI-Driven Medical Second Opinion System (AI-MSOS), a clinical decision support platform developed to address diagnostic inequity in resource-constrained healthcare environments. The document follows the structure recommended by IEEE Standard 830-1998 and draws on ISO/IEC 25010:2023 quality characteristics to define measurable software quality attributes.

Healthcare systems across low- and middle-income countries confront a persistent specialist deficit that directly impairs diagnostic accuracy at primary care facilities. The AI-MSOS responds to this challenge by delivering machine learning–powered diagnostic validation directly to frontline general practitioners, enabling evidence-based clinical reasoning at the point of care without requiring synchronous specialist availability.

1.1 Purpose

This document serves as the authoritative technical reference for all parties involved in the design, development, deployment, validation, and governance of the AI-MSOS. It communicates system intent to software engineers, clinical informaticists, quality assurance teams, regulatory bodies, and institutional stakeholders. The specification delineates the expected behaviour of the system, the constraints within which it operates, and the standards to which it must conform. Any ambiguity in system requirements should be resolved by reference to this document in conjunction with the attached clinical use-case annexes.

1.2 Scope

The AI-MSOS is a web-accessible and mobile-accessible clinical intelligence platform designed to function in environments characterised by limited healthcare infrastructure, intermittent connectivity, and a shortage of specialist physicians. Within the scope of this specification, the system shall perform the following primary functions:

- Accept structured clinical inputs encompassing patient demographics, presenting symptoms, vital signs, laboratory investigation results, and DICOM-formatted medical imaging files
- Apply trained machine learning models to generate ranked differential diagnoses with associated confidence scores and supporting clinical rationale
- Facilitate an asynchronous specialist consultation workflow in which credentialed medical consultants may review, annotate, and validate AI-generated recommendations
- Maintain an immutable, tamper-evident audit trail of all clinical decisions, data access events, and model interactions
- Interoperate with electronic health record systems through HL7 FHIR R4 APIs
- Support offline data capture with automatic background synchronisation upon network reconnection
- Comply with HIPAA, GDPR, and applicable national health data regulations across target deployment regions

1.3 Problem Definition

The World Health Organization estimates a global deficit exceeding 4.3 million healthcare workers, with the most acute shortages concentrated in sub-Saharan Africa and South Asia. In these regions, general practitioners at district and rural facilities routinely encounter clinical presentations that demand specialist

expertise unavailable within reasonable geographic or temporal proximity. Diagnostic error rates in such settings are estimated between 28% and 40%, contributing to inappropriate treatment, avoidable patient harm, and unnecessary utilisation of limited healthcare resources.

Existing telemedicine frameworks address some of these constraints but presuppose stable broadband connectivity and the real-time availability of specialist consultants — conditions that cannot be reliably met across much of the developing world. The AI-MSOS resolves this by operating as an asynchronous, AI-powered diagnostic validation layer capable of functioning under intermittent 2G/3G conditions. By encoding specialist-level clinical reasoning into validated machine learning models, the system delivers meaningful decision support at the point of care, independent of specialist availability or network stability.

1.4 Audience

This specification is written for the following stakeholders, each of whom may use it as a reference for their specific responsibilities:

- General Practitioners and frontline healthcare workers in primary and district-level care facilities, who constitute the primary end users of the diagnostic second opinion workflow
- Medical Specialist Consultants who participate in asynchronous case review and serve as validators within the platform's clinical governance framework
- Hospital Administrators and Healthcare System Managers responsible for institutional deployment and user onboarding
- Software Engineers and System Architects tasked with implementing the platform in accordance with this specification
- Quality Assurance and Clinical Validation Teams conducting system testing and model performance evaluation
- Regulatory and Compliance Officers overseeing Software as a Medical Device (SaMD) certification and data governance
- Public Health Officials and NGO Partners engaged in health system strengthening initiatives across target deployment regions

1.5 Tools to be Used

The platform is implemented using the following technology stack, selected to optimise clinical performance, security, and operability in low-resource environments:

- Frontend: React.js with TypeScript, Tailwind CSS, Progressive Web App (PWA) manifest for offline support
- Backend: Python (FastAPI), Node.js microservices for event-driven workflows
- AI and Machine Learning: TensorFlow 2.x, PyTorch, Hugging Face Transformers (BioBERT, ClinicalBERT)
- Medical Imaging AI: MONAI framework, OpenCV for pre-processing; DICOM standard for imaging interchange
- Database: PostgreSQL 15 (relational data), MongoDB (unstructured clinical notes), Redis (session caching)
- Cloud Infrastructure: AWS HealthLake or GCP Healthcare API, Docker, Kubernetes for orchestration
- Security: OAuth 2.0, OpenID Connect, AES-256 encryption at rest, TLS 1.3 in transit, RBAC
- Interoperability: HL7 FHIR R4 API, DICOM WADO-RS for imaging retrieval

1.6 References

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- ISO/IEC 25010:2023. Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE). International Organization for Standardization.
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1.7 Overview

The remainder of this document is organised as follows. Section 2 provides a Glossary of domain-specific medical and technical terminology together with an Acronyms and Abbreviations table. Section 3 presents the System Model, comprising six architectural and behavioural diagrams: State Transition, UML Class, Sequence, Use Case, Data Flow, and Entity Relationship diagrams. Section 4 specifies Functional Requirements, organised by user class. Section 5 defines Non-Functional Requirements across security, performance, safety, and usability dimensions. Section 6 describes the System Evolution roadmap across four phased delivery milestones. Section 7 provides Detailed Requirement Specifications, including hardware interface constraints, product function summary, complete database schema, user characteristics, and assumptions and dependencies.

2. GLOSSARY

2.1 Definitions

AI Diagnosis Engine. A computational subsystem that applies trained supervised and unsupervised machine learning models to structured patient data in order to generate a ranked differential diagnosis list, each entry annotated with a probabilistic confidence score and a citation of supporting clinical evidence patterns derived from the training corpus.

Clinical Decision Support System (CDSS). Any health information technology application designed to enhance clinician decision-making by providing evidence-based clinical knowledge at the point of care. The AI-MSOS functions as an advanced CDSS by integrating real-time patient data with a validated clinical knowledge base and specialist-reviewed diagnostic logic.

Differential Diagnosis (DDx). A structured clinical methodology in which the clinician or diagnostic system systematically evaluates the full spectrum of medical conditions consistent with a patient's presenting symptoms, signs, and diagnostic findings, and ranks them according to probability. The AI-MSOS produces ranked DDx lists as its primary output artefact.

FHIR (Fast Healthcare Interoperability Resources). An international interoperability standard for healthcare data exchange, maintained by HL7 International. FHIR defines a set of resources that model clinical and administrative data and exposes them through RESTful APIs. The AI-MSOS adopts FHIR R4 as its primary interoperability interface.

Confidence Score. A numerical measure, expressed as a percentage on the interval [0, 100], that quantifies the AI engine's probabilistic certainty that a given diagnosis correctly characterises the patient's clinical presentation. The score is derived from the posterior probability output of the classification model, calibrated against a held-out validation dataset.

DICOM (Digital Imaging and Communications in Medicine). The internationally adopted standard for the management, storage, transmission, and display of medical imaging information and associated patient metadata. The AI-MSOS ingests DICOM-formatted files for AI-assisted radiology analysis and provides WADO-RS compliant image retrieval endpoints.

Software as a Medical Device (SaMD). Software intended to be used for one or more medical purposes that performs those purposes without being part of a hardware medical device, as defined by the International Medical Device Regulators Forum (IMDRF). The AI-MSOS is classified as SaMD and is developed in accordance with FDA and applicable national SaMD regulatory guidance.

Federated Learning. A distributed machine learning paradigm in which model training occurs locally on data residing at individual healthcare institutions, with only model weight updates transmitted to a central aggregation server. This approach allows the AI-MSOS to improve its diagnostic models over time without centralising sensitive patient data, thereby preserving data sovereignty and regulatory compliance.

2.2 Acronyms and Abbreviations

Acronym	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CDSS	Clinical Decision Support System
CBC	Complete Blood Count
DDx	Differential Diagnosis
DICOM	Digital Imaging and Communications in Medicine
EHR	Electronic Health Record
FHIR	Fast Healthcare Interoperability Resources
GP	General Practitioner
GDPR	General Data Protection Regulation
HL7	Health Level Seven International
HIPAA	Health Insurance Portability and Accountability Act
ICD-10	International Classification of Diseases, 10th Revision
IMDRF	International Medical Device Regulators Forum
LMIC	Low- and Middle-Income Country
ML	Machine Learning
MFA	Multi-Factor Authentication
MONAI	Medical Open Network for AI
NLP	Natural Language Processing
PHI	Protected Health Information
PWA	Progressive Web Application
RBAC	Role-Based Access Control
REST	Representational State Transfer
SaMD	Software as a Medical Device
SRS	Software Requirements Specification
TOTP	Time-Based One-Time Password
WHO	World Health Organization

3. SYSTEM MODEL

This section presents six architectural and behavioural models that collectively characterise the structure, dynamics, and data organisation of the AI-MSOS. Each diagram is accompanied by a descriptive narrative. All diagrams were produced programmatically and reflect the logical design of the system.

3.1 State Transition Diagram

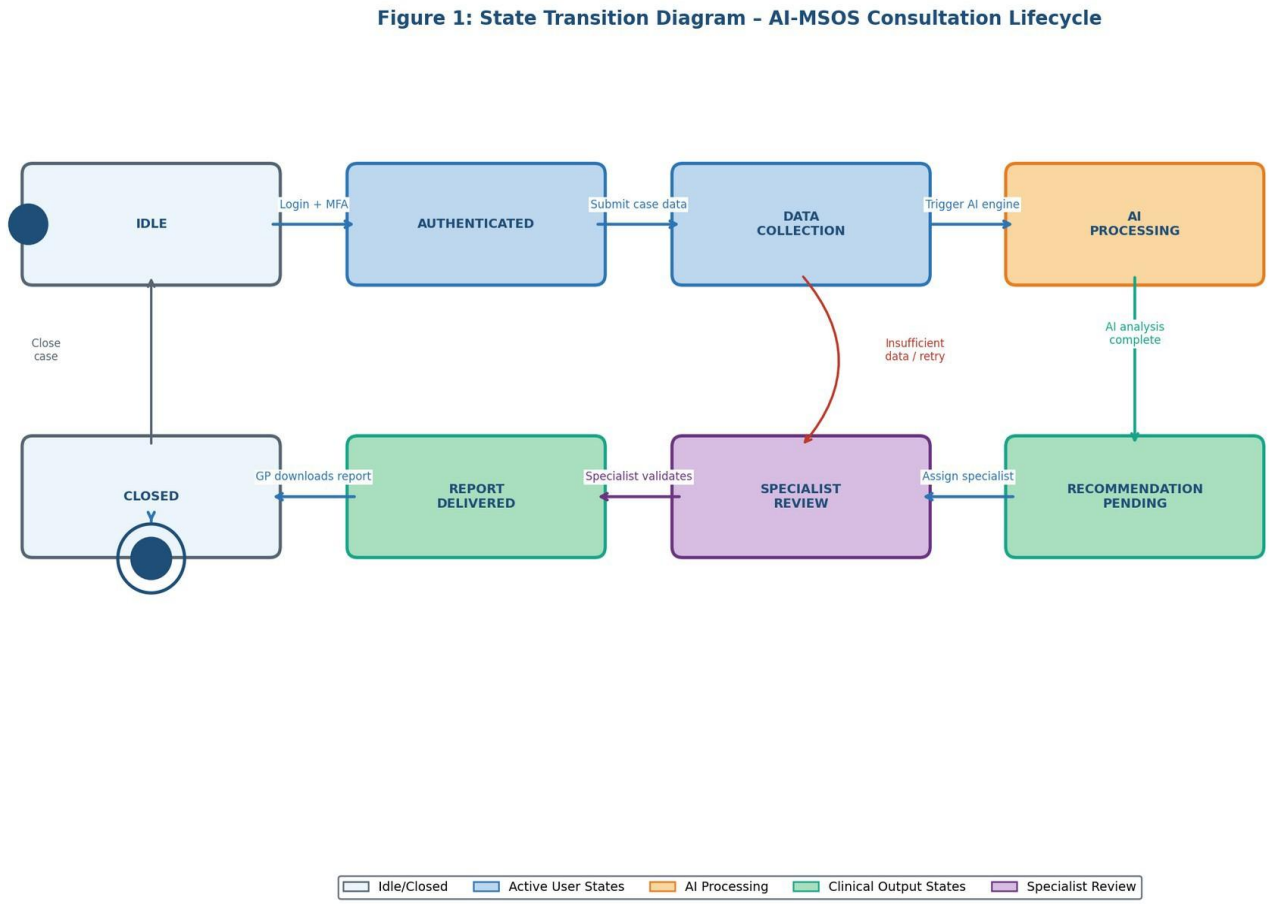

Figure 1: State Transition Diagram — AI-MSOS Diagnostic Consultation Lifecycle

The State Transition Diagram presented in Figure 1 models the lifecycle of a diagnostic consultation request from initiation through to closure. The system commences in the Idle State, awaiting user interaction. Upon successful credential validation and multi-factor authentication, it transitions to the Authenticated State. Patient data entry and submission advance the workflow to the Data Collection State, which captures demographic information, symptoms, vital signs, laboratory results, and optional imaging. Once the case is submitted, the system enters the AI Processing State, wherein the Diagnosis Engine applies trained models to generate a ranked differential diagnosis report. The resulting output is held in the Recommendation Pending State pending either autonomous delivery to the requesting clinician or allocation to a specialist for validation. The Specialist Review State accommodates cases escalated for expert annotation. Upon completion, the Final Report Delivered State makes the validated clinical report available for download. A re-submission transition from Specialist Review back to Data Collection accommodates cases in which submitted information is deemed insufficient for a reliable diagnosis. The workflow concludes in the Closed State, from which the case enters long-term archival storage.

3.2 UML Class Diagram

Figure 2: UML Class Diagram - AI-MSOS Data Model

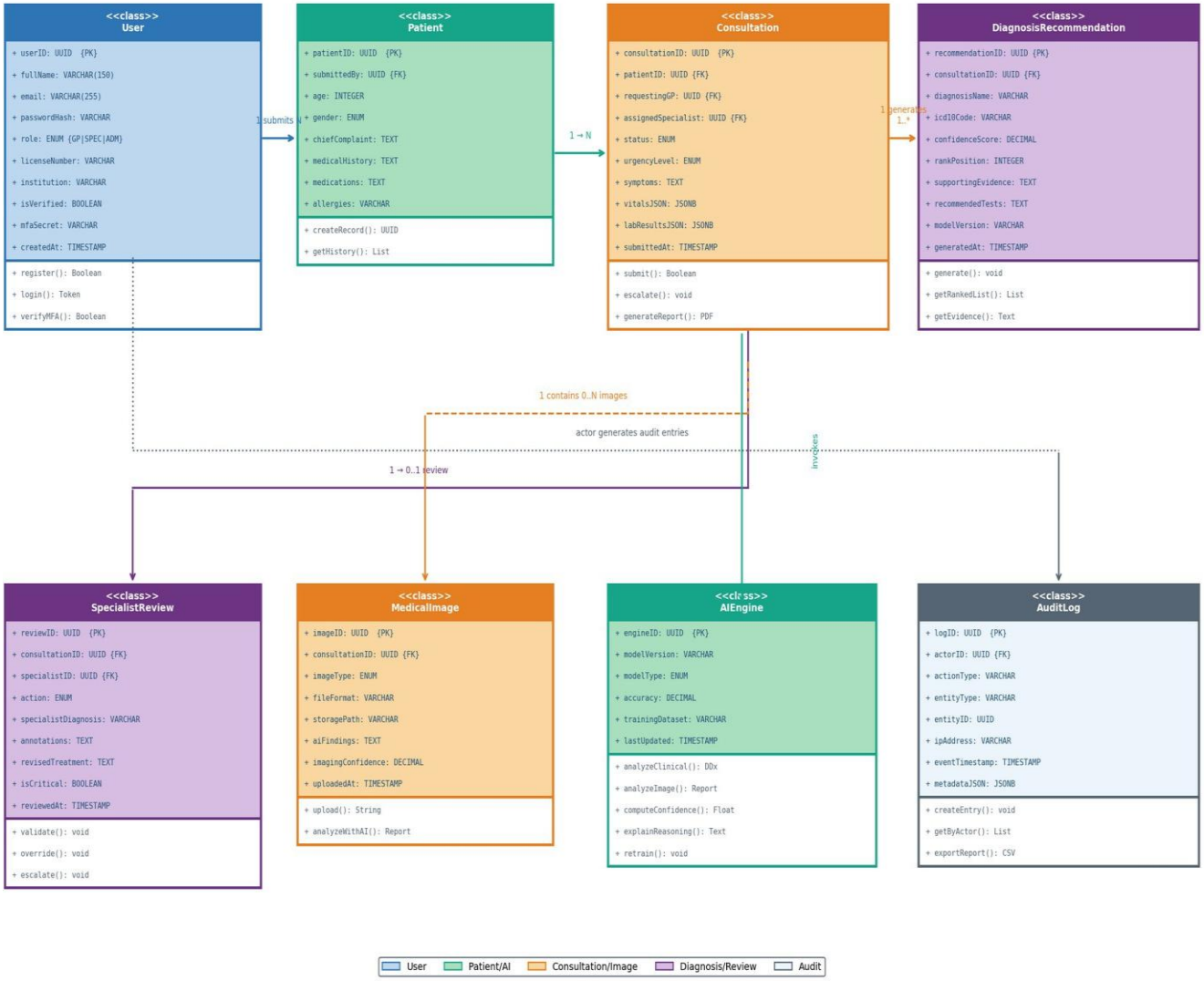

Figure 2: UML Class Diagram — AI-MSOS Data Model and Object Relationships

Figure 2 presents the UML Class Diagram, which delineates the eight principal domain entities of the AI-MSOS and the associations between them. The User class serves as the base entity for all authenticated actors, with role enumeration distinguishing General Practitioners, Specialist Consultants, and System Administrators. A User in the GP role may submit multiple Patient records, each of which may be associated with multiple Consultation instances. Each Consultation aggregates zero or more MedicalImage objects and triggers the AIEngine class to generate one or more DiagnosisRecommendation objects, each characterised by an ICD-10 code, a confidence score, and a ranked position within the differential diagnosis list. Consultations escalated for specialist review generate a SpecialistReview object, which encapsulates the consultant's action, annotations, and any revised treatment plan. Every system interaction produces an AuditLog entry linked to the acting User, ensuring complete accountability for all clinical decisions and data access events. Method signatures are included for each class to communicate the principal behaviours expected during implementation.

3.3 Sequence Diagram

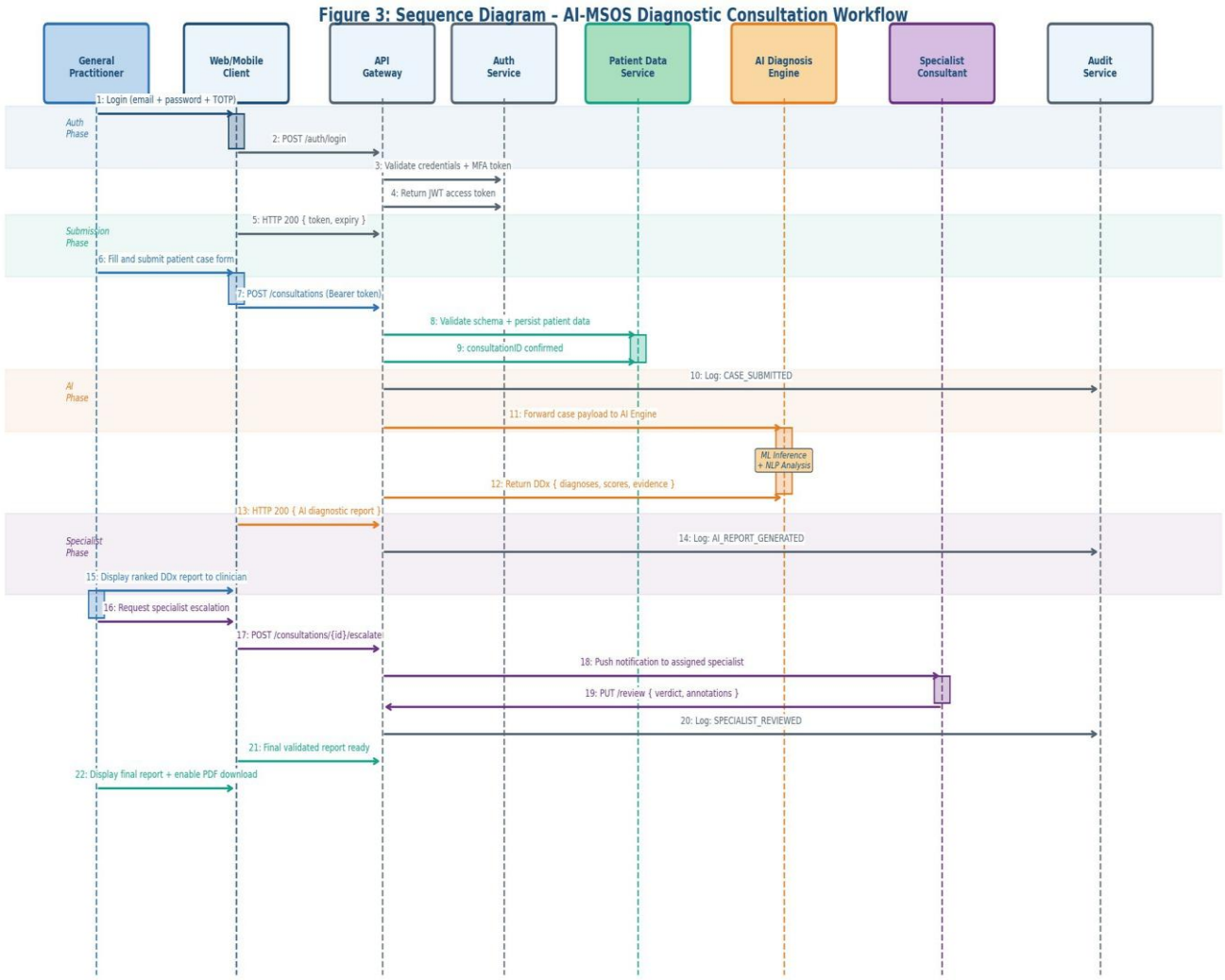

Figure 3: Sequence Diagram — AI-MSOS Consultation Workflow Interaction

Figure 3 illustrates the temporal sequence of interactions between system actors and internal components during a complete diagnostic consultation cycle, organised across four colour-coded phases. In the Authentication Phase, the General Practitioner submits credentials and a TOTP token through the web or mobile client; the API Gateway routes the request to the Auth Service, which validates the credentials, issues a JWT access token, and returns it to the client. During the Submission Phase, the GP populates and submits the patient case form; the API Gateway forwards the validated payload to the Patient Data Service for persistence, and a CASE_SUBMITTED event is written to the Audit Service. The AI Processing Phase begins when the Gateway dispatches the case payload to the AI Diagnosis Engine, which performs NLP analysis and model inference before returning a ranked DDx report with confidence scores; an AI_REPORT_GENERATED event is logged. The Specialist Phase is initiated when the GP requests escalation; the system notifies an assigned specialist, who submits a review verdict and annotations; a SPECIALIST_REVIEWED event is committed to the audit trail, and the finalised report is returned to the requesting clinician. All inter-service communication is authenticated through JWT Bearer tokens and encrypted via TLS 1.3.

3.4 Use Case Diagram

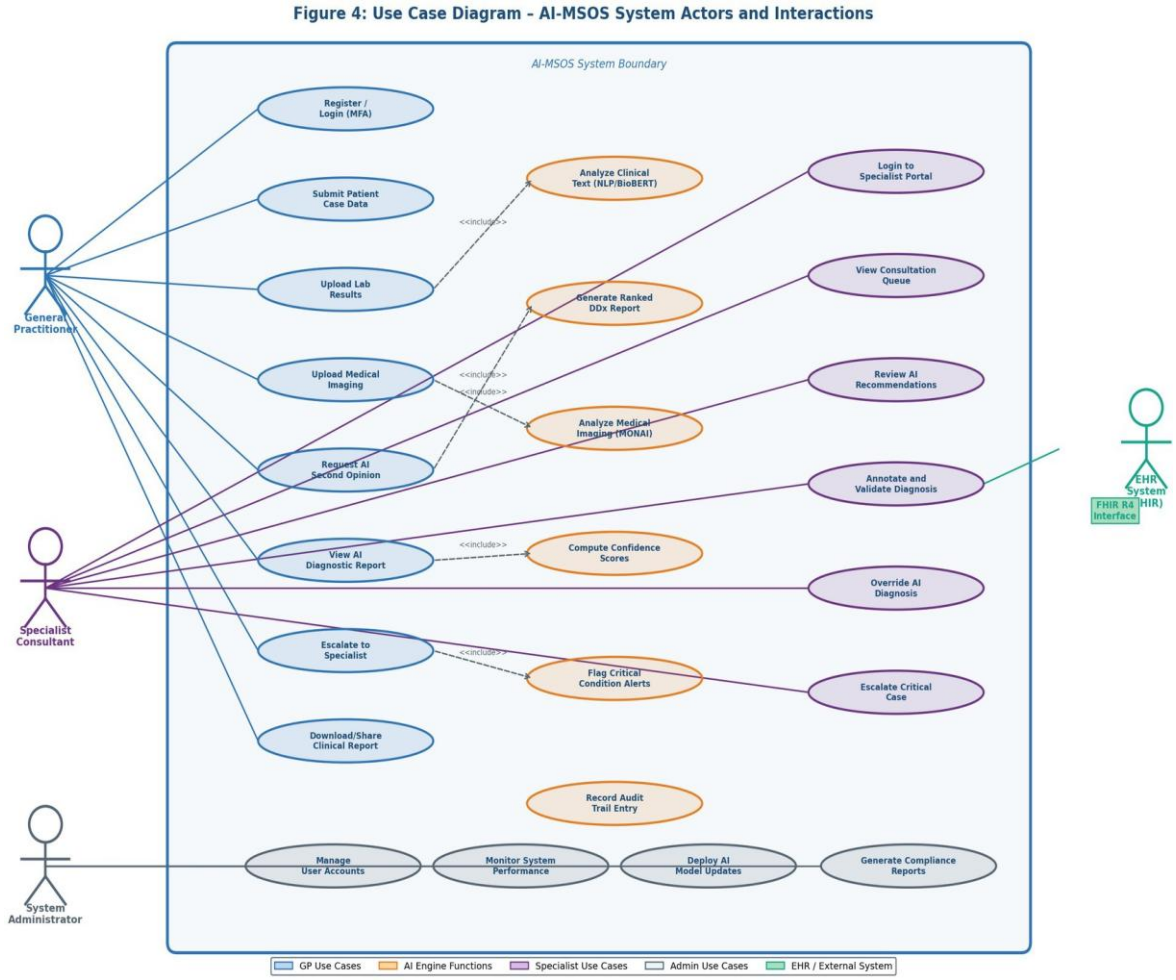

Figure 4: Use Case Diagram — AI-MSOS System Actors and Interactions

Figure 4 depicts the Use Case Diagram, which identifies four actors interacting with the AI-MSOS system boundary. The General Practitioner actor engages with eight use cases spanning the complete consultation workflow: user registration and login with MFA, patient case submission, laboratory result upload, medical imaging upload, AI second opinion request, diagnostic report review, specialist escalation, and report download. The Specialist Consultant actor participates in six use cases: portal login, consultation queue review, AI recommendation examination, diagnostic annotation and validation, AI diagnosis override, and critical case escalation. The System Administrator actor manages platform operations through four use cases: user account management, system performance monitoring, AI model update deployment, and compliance report generation. The EHR System external actor interacts with the platform via the FHIR R4 API for bidirectional patient data exchange. Include relationships from GP use cases to AI Engine functions highlight the system's internal delegation of clinical analysis to its ML subsystems.

3.5 Data Flow Diagram

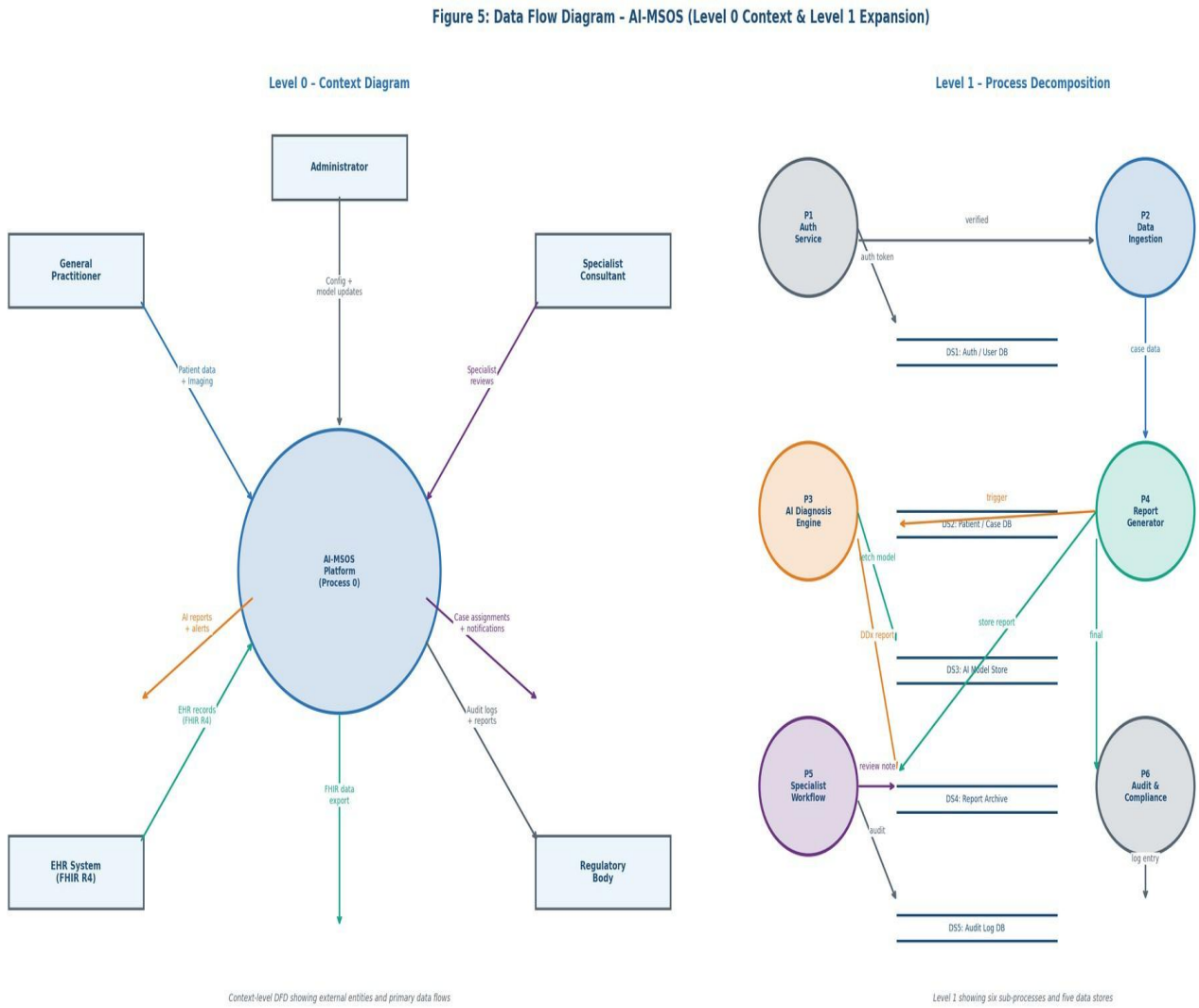

Figure 5: Data Flow Diagram — Level 0 Context and Level 1 Process Decomposition

Figure 5 presents the Data Flow Diagram across two levels of abstraction. The Level 0 Context Diagram represents the AI-MSOS as a single processing entity and maps its interactions with five external entities: General Practitioner, Specialist Consultant, EHR System, Regulatory Body, and Administrator. Inputs to the system include patient clinical data with imaging from the GP, specialist review submissions, FHIR-formatted EHR records, and administrative configuration updates. Outputs include AI diagnostic reports and critical alerts to the GP, case assignment notifications to the Specialist, compliance and audit logs to the Regulatory Body, and validated FHIR exports to the EHR System. The Level 1 Decomposition diagram expands the central process into six sub-processes: P1 Auth Service, P2 Data Ingestion, P3 AI Diagnosis Engine, P4 Report Generator, P5 Specialist Workflow, and P6 Audit and Compliance. Five data stores support these processes: DS1 Auth/User DB, DS2 Patient/Case DB, DS3 AI Model Store, DS4 Report Archive, and DS5 Audit Log DB. Data flows between processes and stores are annotated to clarify the direction and content of each exchange.

3.6 Entity Relationship Diagram

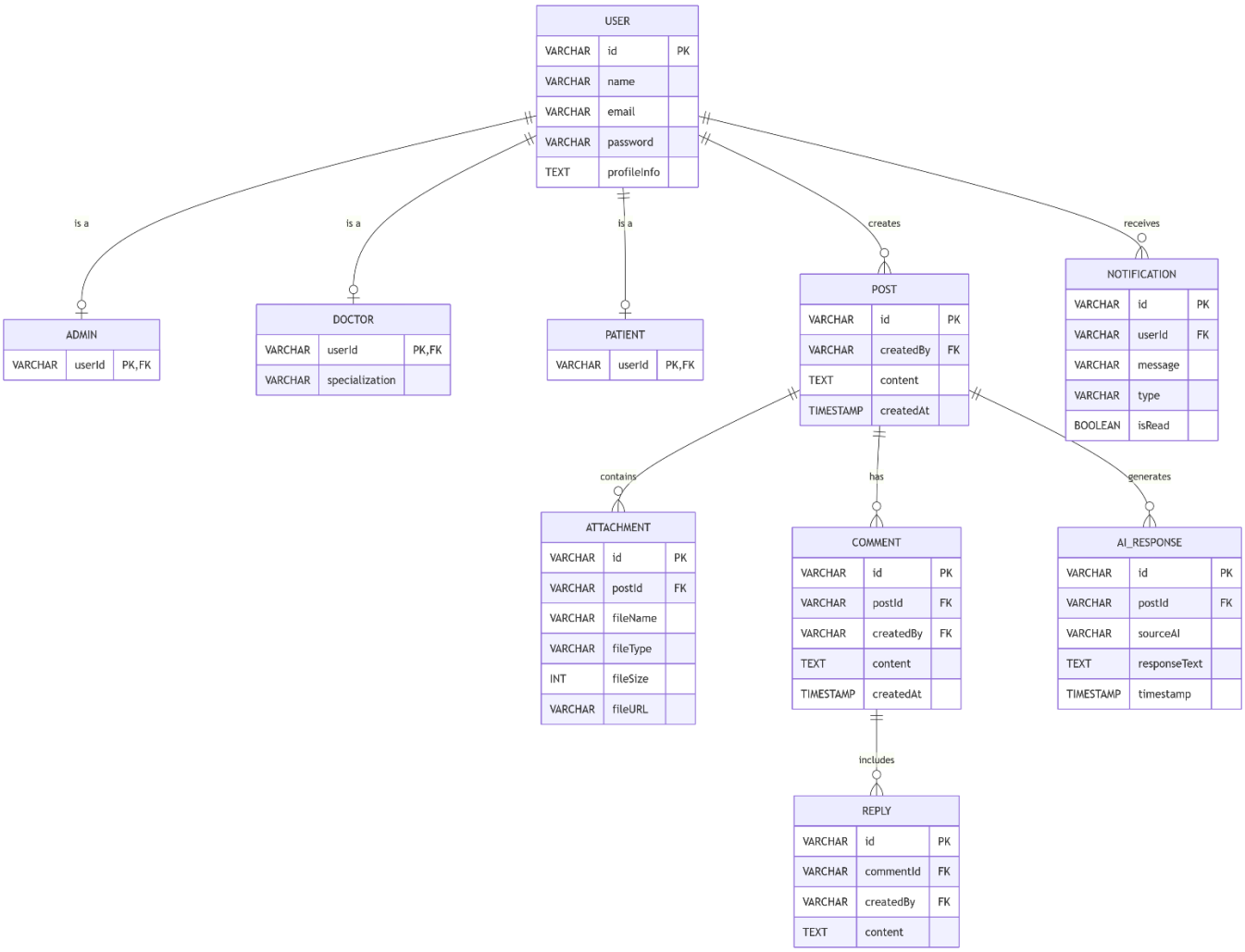

Figure 6: Entity Relationship Diagram — AI-MSOS Database Schema

Figure 6 presents the Entity Relationship Diagram for the AI-MSOS relational database schema. The MediSecond database schema employs a Class Table Inheritance (CTI) pattern to manage polymorphic user associations. The core data model revolves around the USER entity, which establishes global authentication credentials and demographic data. Sub-type entities (ADMIN, DOCTOR, and PATIENT) maintain a strictly 1:1 relationship with the base USER table, acting as specialized extensions that hold role-specific attributes (e.g., the specialization attribute within the DOCTOR entity). The primary transactional artifact of the system is the POST entity, representing a medical case submission. It maintains a mandatory many-to-one relationship with the USER entity (specifically instantiated by a Patient). The POST entity acts as a composite root, enforcing a one-to-many lifecycle dependency with three distinct child entities: ATTACHMENT (handling associated medical imagery or documents via file URL pointers), COMMENT (capturing clinical feedback from human actors), and AI_RESPONSE (storing deterministic triage data and diagnostic perspectives generated by the AI module). Hierarchical interactions are supported through the REPLY entity, which maintains a many-to-one foreign key constraint against a parent COMMENT, allowing for deep-threaded clinical discussions. Finally, system-wide asynchronous events are managed via the NOTIFICATION entity, which maps back to the USER base class in a many-to-one relationship, utilizing a boolean isRead flag to track state persistence across the user interface.

4. FUNCTIONAL REQUIREMENTS

Functional requirements are organised by user class. Each requirement carries a unique identifier, a title, and a description specifying the expected system behaviour. Requirements at this level represent contractual obligations of the system to its users and must be verifiable through defined acceptance tests.

4.1 User Class 1 — General Practitioners and Frontline Healthcare Workers

❖ **Functional Requirement FR-GP-01**

TITLE: User Registration and Account Creation
DESC: General Practitioners and frontline healthcare workers may register on the platform by providing their full legal name, professional designation, medical licence number, affiliated institution, country of practice, primary contact email address, and a password satisfying the platform's complexity policy. Upon submission, the system dispatches a verification email containing a tokenised activation link valid for 24 hours. Account activation additionally requires approval from a designated institutional administrator before the account is granted access to clinical functions.

❖ **Functional Requirement FR-GP-02**

TITLE: Secure Login with Multi-Factor Authentication
DESC: Registered GPs authenticate using their registered email address and password. The system mandates multi-factor authentication for all clinical sessions, accepting a time-based one-time password (TOTP) generated by a compatible authenticator application or delivered via SMS. Following five consecutive failed login attempts within a 10-minute window, the account is locked for 15 minutes and the registered email address is notified.

❖ **Functional Requirement FR-GP-03**

TITLE: Patient Case Submission
DESC: Authenticated GPs may initiate a new consultation case by entering the patient's age, biological sex, body weight, blood group, chief complaint, detailed symptom narrative, symptom duration, vital signs (temperature, systolic and diastolic blood pressure, pulse rate, respiratory rate, and peripheral oxygen saturation), relevant medical and surgical history, current medications with dosage, and known allergies. The system validates completeness of mandatory fields before accepting a submission and assigns a unique consultation identifier upon successful creation.

❖ **Functional Requirement FR-GP-04**

TITLE: Laboratory Results Upload
DESC: GPs may attach structured laboratory investigation results to a consultation, including but not limited to complete blood count, comprehensive metabolic panel, liver function tests, renal function tests, coagulation profile, urinalysis, and microbiology culture results. Results may be entered via structured form fields or uploaded as PDF, CSV, or image files. The system applies optical character recognition to image-based uploads and structures extracted values for consumption by the AI Diagnosis Engine.

❖ **Functional Requirement FR-GP-05**

TITLE: Medical Imaging Upload
DESC: GPs may upload medical imaging files in DICOM format or as JPEG and PNG images, covering modalities including plain radiography, computed tomography, magnetic resonance imaging, and ultrasound. The system validates file integrity upon upload and routes accepted files to the AI imaging analysis module. Imaging findings are integrated into the diagnostic report alongside text-based clinical analysis.

- **Functional Requirement FR-GP-06**

TITLE: AI Diagnostic Second Opinion Request
DESC: Following the submission of a patient case, GPs may request an AI-generated diagnostic second opinion. The system processes the complete case payload through the AI Diagnosis Engine and delivers a ranked differential diagnosis report within 60 seconds under normal operating load. The report includes a minimum of five differential diagnoses ranked by confidence score, supporting clinical evidence for each entry, red-flag alerts for conditions indicating immediate clinical urgency, and recommended follow-up investigations.

- **Functional Requirement FR-GP-07**

TITLE: View and Interpret AI Recommendations
DESC: GPs may access the AI-generated diagnostic report through an interactive interface presenting ranked differential diagnoses with probability percentages, an explainability panel summarising the clinical features that most influenced each recommendation, annotated red-flag alerts for critical conditions, recommended investigations with clinical rationale, and contextual links to relevant clinical practice guidelines.

- **Functional Requirement FR-GP-08**

TITLE: Request Specialist Consultation
DESC: For cases in which the AI recommendations are inconclusive or the clinical presentation is of high acuity, GPs may escalate the consultation to a human specialist consultant. The GP selects the appropriate medical specialty from a pre-defined list, appends a consultation note, and submits the request. The system notifies eligible specialists and tracks response time against service level targets of four hours for routine cases and 30 minutes for critical cases.

- **Functional Requirement FR-GP-09**

TITLE: Case History and Patient Record Access
DESC: GPs may search and retrieve a complete history of all consultations submitted under their account, including current workflow status, AI recommendations generated, specialist annotations received, and follow-up outcome data entered subsequently. Patient records may be updated with post-consultation findings to support longitudinal care tracking and model retraining feedback cycles.

- **Functional Requirement FR-GP-10**

TITLE: Clinical Report Download and Sharing
DESC: GPs may download a structured PDF clinical report consolidating the patient case summary, AI-generated recommendations, specialist annotations, and a clinical disclaimer. Reports may alternatively be exported directly to a connected EHR system via the FHIR R4 API or shared with the patient or referral facility through a time-limited, encrypted access link.

4.2 User Class 2 — Medical Specialist Consultants

- **Functional Requirement FR-SP-01**

TITLE: Specialist Account Registration and Credentialing
DESC: Medical specialist consultants register by providing their full name, primary clinical specialty, subspecialty where applicable, years of postgraduate clinical experience, affiliated institution, country of practice, and medical licence or registration number. Submitted credentials undergo a manual verification process conducted by platform administrators, during which the account remains inactive. Upon

successful verification, the system activates the specialist account and grants access to the consultation review portal.

❖ **Functional Requirement FR-SP-02**

TITLE: View Assigned Consultation Queue
DESC: Authenticated specialists access a prioritised queue of consultation requests allocated to their declared specialty. Each queue entry displays the case urgency classification, a patient clinical summary, the requesting institution and GP identifier, the AI-generated DDx report, and the elapsed time since submission. The queue supports filtering by urgency level, specialty, submission date, and review status.

❖ **Functional Requirement FR-SP-03**

TITLE: Review AI-Generated Diagnostic Recommendations
DESC: Specialists may examine the complete AI diagnostic report for each assigned case, including the ranked DDx list with confidence scores, the clinical evidence cited in support of each recommendation, and AI-generated imaging analysis findings where applicable. The interface presents the AI reasoning pathway in a structured explainability panel, enabling the specialist to assess the clinical validity of the model's conclusions before reaching an independent judgement.

❖ **Functional Requirement FR-SP-04**

TITLE: Annotate, Validate, or Override AI Diagnosis
DESC: Specialists may accept the AI-generated diagnosis as presented, partially modify the ranked list by reordering or supplementing entries, or override the AI recommendation entirely and substitute a specialist diagnosis. All actions are accompanied by mandatory free-text annotations. Modifications are persisted with the specialist's user identifier and a UTC timestamp and are surfaced to the requesting GP alongside the original AI report.

❖ **Functional Requirement FR-SP-05**

TITLE: Escalate Critical Cases
DESC: Specialists may flag a case as clinically critical if the presenting data indicates an immediately life-threatening condition. Such a flag triggers a real-time push notification to the requesting GP and alerts the relevant institutional administrator. The critical flag is recorded in the SpecialistReview entity and is prominently displayed in the final report delivered to the clinician.

❖ **Functional Requirement FR-SP-06**

TITLE: Secure Communication with General Practitioner
DESC: Specialists may initiate a secure, in-platform message thread with the requesting GP, including file attachments, annotated imaging, and voice note recordings. All communications are encrypted end-to-end, logged in association with the corresponding consultation record, and are retained as part of the case audit trail for the mandatory retention period.

4.3 User Class 3 — System Administrators

❖ **Functional Requirement FR-ADM-01**

TITLE: User Account Management
DESC: Administrators may create, modify, deactivate, and permanently delete user accounts for GPs, Specialist Consultants, and other administrative users. The administrator portal supports role assignment, institution affiliation management, and manual credential verification workflows for specialist accounts. Bulk user provisioning is supported via structured CSV import with validation.

❖ Functional Requirement FR-ADM-02

TITLE: System Monitoring and Performance Dashboard

DESC: Administrators have access to a real-time operational dashboard displaying active concurrent sessions, consultation throughput volumes, AI engine inference latency distributions, system resource utilisation metrics, error rates by service component, and platform uptime. Automated threshold-based alerts notify administrators via email and in-application notification when defined performance thresholds are breached.

❖ Functional Requirement FR-ADM-03

TITLE: AI Model Management and Version Control

DESC: Administrators may initiate new AI model deployments through a controlled release pipeline that includes staging environment validation, clinical performance benchmarking against held-out test datasets, and approval gating before production promotion. A complete version history of deployed models is maintained, and rollback to any prior verified version is supported without service interruption. Model performance metrics including accuracy, AUC-ROC, sensitivity, and specificity are surfaced through the administration dashboard.

❖ Functional Requirement FR-ADM-04

TITLE: Audit Log Review and Compliance Reporting

DESC: Administrators may query the immutable audit log using filters on actor identity, action type, entity type, affected entity identifier, date range, and IP address. Compliance reports summarising system activity, data access patterns, and AI recommendation accuracy across defined reporting periods may be generated in PDF and CSV formats for regulatory submission to applicable health data authorities.

5. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality attributes that the AI-MSOS must exhibit in addition to its functional capabilities. These requirements are measurable, testable, and traceable to the ISO/IEC 25010:2023 quality model. Failure to satisfy any of the requirements in this section constitutes a non-conformance that must be resolved before production deployment.

5.1 Security Requirements

- All Protected Health Information (PHI) stored in the system's databases and file storage infrastructure shall be encrypted using AES-256 in compliance with HIPAA Security Rule §164.312(a)(2)(iv)
- All data transmitted between client applications, the API Gateway, and internal microservices shall be encrypted using TLS 1.3; support for TLS 1.1 and earlier versions shall be explicitly disabled
- Multi-factor authentication using a TOTP-based mechanism shall be mandatory for all user accounts possessing access to clinical data; there shall be no exception or bypass pathway
- Access to all system resources shall be governed by a role-based access control (RBAC) model enforced at the API Gateway layer, ensuring that users may access only the data objects and operations explicitly authorised for their assigned role
- User passwords shall contain a minimum of ten characters and shall satisfy complexity requirements including at least one uppercase letter, one lowercase letter, one numeral, and one special character
- All authentication events, data retrieval operations, AI model invocations, specialist annotations, and administrative actions shall be written to an append-only, cryptographically verifiable audit log
- The system shall comply with HIPAA Security Rule, GDPR Article 25 (data protection by design and by default), and applicable national health data legislation in each deployment country
- Automated vulnerability scanning of all platform components shall be conducted at minimum monthly, and penetration testing by a qualified third party shall be performed semi-annually

5.2 Performance Requirements

- The AI Diagnosis Engine shall return a complete ranked differential diagnosis report within 60 seconds of case submission under standard operating conditions, defined as up to 1,000 concurrent analysis requests
- AI-assisted analysis of a standard-resolution DICOM imaging series (up to 512 × 512 pixels, 100 slices) shall complete within three minutes
- The platform shall sustain a minimum of 5,000 concurrent authenticated user sessions without measurable degradation in response time for standard REST API operations
- All REST API read operations shall respond within 200 milliseconds and write operations within 500 milliseconds at the 95th percentile under peak load, as measured by continuous performance monitoring infrastructure
- The platform shall maintain a minimum monthly uptime of 99.5%, equating to a maximum of 3.65 hours of unplanned downtime per month, with planned maintenance windows communicated at least 72 hours in advance

- The Progressive Web App shall achieve full interactive readiness within three seconds on a simulated 3G network connection (1.6 Mbps downlink) as measured by Lighthouse performance scoring
- The PWA shall support offline-first case submission queuing, automatically synchronising queued cases to the server within 30 seconds of network reconnection

5.3 Safety Requirements

- Every AI-generated diagnostic recommendation presented to a clinician shall be accompanied by a mandatory regulatory disclaimer stating that the output constitutes clinical decision support and does not replace the independent clinical judgement of a licensed healthcare professional
- Any DiagnosisRecommendation with a confidence score below 60% shall be flagged with a visual indicator prompting the GP to consider specialist escalation before acting on the recommendation
- The system shall include a set of pre-defined critical condition detectors, encompassing indicators for sepsis, acute myocardial infarction, stroke, pulmonary embolism, and meningitis, which trigger real-time push notifications to the GP and on-call specialist whenever a case presentation meets defined threshold criteria
- All AI model updates promoted to production shall have been validated on a pre-defined clinical holdout dataset demonstrating a minimum sensitivity of 85% and specificity of 80% across the system's core diagnostic categories
- The platform shall implement a structured feedback mechanism through which GPs and Specialists may flag AI recommendations as clinically incorrect; flagged cases shall feed into a model retraining review pipeline without direct modification of production model weights
- Patient data shall be backed up at intervals not exceeding six hours, with point-in-time recovery capability maintained for a rolling period of at least 30 days

5.4 Usability Requirements

- The user interface shall conform to WCAG 2.1 Level AA accessibility guidelines across all primary user-facing screens
- The platform shall support multilingual interfaces in a minimum of five languages at initial release: English, French, Arabic, Swahili, and Portuguese, with additional languages configurable through the administration portal
- The mobile application shall operate correctly on Android devices running Android 8.0 (API level 26) or later with a minimum of 2 GB RAM, and on iOS devices running iOS 13.0 or later
- A newly registered GP user shall be able to complete their first full consultation submission without external guidance within ten minutes of account activation, as validated through usability testing with representative users in target deployment regions
- The system shall provide contextual in-application guidance, including field-level tooltips, workflow progress indicators, and an integrated help centre with searchable articles, within all primary clinical workflow screens

6. SYSTEM EVOLUTION

The AI-MSOS is architected as an iteratively deliverable platform following an agile release model. Development proceeds across four clearly demarcated phases, each building on validated functionality from the preceding phase while maintaining backward compatibility with deployed components and integrated EHR systems. The phased approach reduces clinical risk by limiting the surface area of change introduced at any single deployment event.

Phase 1 — Minimum Viable Product (Months 1–4)

The MVP phase delivers the foundational capabilities required to support the core diagnostic second opinion workflow in a controlled pilot environment:

- User authentication system with RBAC, covering GP and Administrator roles, with MFA enforcement
- Structured patient case submission form with mandatory field validation and consultation identifier assignment
- Basic DDx report generation with confidence scores and supporting evidence
- Foundational audit logging covering authentication and case submission events

Phase 2 — Clinical Intelligence Enhancement (Months 5–8)

Phase 2 extends the platform's clinical capabilities and introduces the specialist consultation workflow:

- Specialist Consultant portal with case queue management, review, annotation, and validation tools
- Text-based AI Diagnosis Engine integrated with a BioBERT-derived clinical NLP model for common conditions across ten primary disease categories
- Medical imaging upload with AI-assisted radiology analysis for plain radiography and basic CT series using the MONAI framework
- Structured laboratory result parser with automated value extraction and AI integration
- Critical condition alerting subsystem with push notifications for defined emergency presentations
- FHIR R4 API integration enabling bidirectional patient data exchange with partner EHR systems
- Multilingual interface support for English, French, and Swahili
- PDF report generation and direct download

Phase 3 — Advanced AI and Regional Deployment (Months 9–14)

Phase 3 extends model capabilities and optimises the platform for field deployment in low-resource settings:

- Extended imaging AI coverage supporting MRI and ultrasound analysis
- Unstructured clinical note parsing using NLP pipelines trained on clinical corpora
- Offline-first Progressive Web App with background synchronisation and service worker caching
- Federated learning infrastructure enabling privacy-preserving model improvement across participating institutions
- AI explainability dashboard presenting SHAP and LIME visualisations of model reasoning pathways

- Integration adapters for national health information systems in primary target deployment countries

Phase 4 — Scale, Compliance, and Optimisation (Months 15–20)

Phase 4 matures the platform toward regulatory submission readiness and enterprise-scale deployment:

- SaMD regulatory documentation package aligned with FDA AI/ML guidance and applicable national equivalents
- Predictive analytics module for public health trend monitoring and outbreak signal detection
- Voice-driven clinical note dictation interface powered by medical-domain speech recognition
- Automated AI model retraining pipeline triggered by performance degradation signals from production monitoring
- Multi-tenant architecture supporting simultaneous deployment across multiple independent health system tenants
- Longitudinal outcome tracking with retrospective diagnostic accuracy analysis against confirmed patient outcomes

7. REQUIREMENT SPECIFICATION

7.1 Product Perspective

The AI-MSOS is a standalone Software as a Medical Device platform that may be deployed as a cloud-hosted multi-tenant service or as an on-premise installation for health systems with data sovereignty constraints. It operates as a clinical decision support overlay that augments existing healthcare workflows without supplanting established diagnostic protocols. External system integration is achieved through standardised APIs, preserving institutional EHR investment while extending diagnostic capability to resource-constrained settings.

7.1.1 Hardware Interface

- Desktop and laptop computers operating any modern standards-compliant web browser (Chrome 90+, Firefox 88+, Edge 90+, Safari 14+) with a minimum display resolution of 1280 × 720 pixels
- Mobile devices running Android 8.0 (API level 26) or later, or iOS 13.0 or later, accessing the platform through the PWA or native application
- Medical tablet devices deployed in clinical settings meeting the minimum mobile operating system requirements above
- Radiology workstations with DICOM viewer integration operating at a minimum display resolution of 1920 × 1080 pixels for imaging review functions
- On-premise server nodes requiring a minimum of 16 CPU cores, 64 GB RAM, and 2 TB NVMe SSD storage per node; GPU-accelerated hardware (minimum NVIDIA T4) is required for local AI inference workloads

7.1.2 Memory and Storage Constraints

- Client-side PWA storage: maximum 50 MB allocated for offline case data caching and submission queuing
- Per-consultation storage footprint: estimated 2–5 MB for text and structured data; up to 500 MB for consultations including full DICOM imaging series
- System-level storage provisioning: minimum 10 TB initial capacity with auto-scaling enabled; storage consumption must be monitored with automated alerting at 75% utilisation
- Audit log retention: a minimum of seven years, conforming to medical record retention standards applicable in the majority of target deployment jurisdictions

7.2 Product Function

A summary of functional capabilities by user class is presented below. Unauthenticated visitors may access only the platform's public landing page, registration form, and publicly available help documentation. No clinical functionality is accessible prior to authentication.

- General Practitioners (Authenticated): submit and manage patient consultation cases; request and view AI diagnostic second opinions; upload laboratory results and medical imaging; escalate cases to specialist review; communicate with assigned specialists; download and export clinical reports

- Specialist Consultants (Authenticated): view and manage a prioritised consultation queue; review AI-generated diagnostic recommendations and imaging analysis; annotate, validate, or override AI diagnoses; escalate critical cases; communicate with requesting GPs
- System Administrators (Authenticated): create and manage user accounts across all roles; monitor platform performance and AI model metrics; govern model update deployments; generate regulatory compliance and audit reports

7.3 Database Requirement

The AI-MSOS employs a hybrid persistence architecture combining a relational database management system (PostgreSQL 15) for structured clinical and administrative data with a document store (MongoDB) for unstructured clinical notes. The following tables define the core relational schema. All UUID primary keys are generated using the UUID v4 specification. Timestamps are stored in UTC.

Table 1: USER

Attribute	Data Type	Constraint	Description
user_id	UUID	PK, NOT NULL	Auto-generated unique identifier
full_name	VARCHAR(150)	NOT NULL	Legal full name
email	VARCHAR(255)	UNIQUE, NOT NULL	Primary login identifier
password_hash	VARCHAR(512)	NOT NULL	Bcrypt-hashed password
role	ENUM	NOT NULL	Values: GP, SPECIALIST, ADMIN
license_number	VARCHAR(100)	NOT NULL	Medical licence number
institution	VARCHAR(255)	NOT NULL	Affiliated hospital or clinic
country	VARCHAR(100)	NOT NULL	Country of practice
specialty	VARCHAR(150)	NULLABLE	Required for SPECIALIST role only
is_verified	BOOLEAN	NOT NULL, DEFAULT FALSE	Account verification status
mfa_secret	VARCHAR(255)	NOT NULL	TOTP shared secret
created_at	TIMESTAMP WITH TIME ZONE	NOT NULL	Record creation timestamp (UTC)
last_login	TIMESTAMP WITH TIME ZONE	NULLABLE	Most recent authenticated session

Table 2: PATIENT

Attribute	Data Type	Constraint	Description
patient_id	UUID	PK, NOT NULL	Unique patient record identifier

Attribute	Data Type	Constraint	Description
submitted_by	UUID	FK → User.user_id, NOT NULL	Submitting GP identifier
age	INTEGER	NOT NULL	Patient age in completed years
gender	ENUM	NOT NULL	Values: MALE, FEMALE, OTHER
weight_kg	DECIMAL(5,2)	NULLABLE	Weight in kilograms
blood_type	VARCHAR(5)	NULLABLE	ABO/Rhesus blood group
chief_complaint	TEXT	NOT NULL	Primary presenting complaint
medical_history	TEXT	NULLABLE	Past medical and surgical history
current_medications	TEXT	NULLABLE	Active medications with dosage
allergies	VARCHAR(500)	NULLABLE	Known drug and environmental allergies
created_at	TIMESTAMP WITH TIME ZONE	NOT NULL	Record creation timestamp (UTC)

Table 3: CONSULTATION

Attribute	Data Type	Constraint	Description
consultation_id	UUID	PK, NOT NULL	Unique consultation identifier
patient_id	UUID	FK → Patient.patient_id, NOT NULL	Associated patient record
requesting_gp	UUID	FK → User.user_id, NOT NULL	Submitting GP identifier
assigned_specialist	UUID	FK → User.user_id, NULLABLE	Assigned specialist identifier
status	ENUM	NOT NULL	PENDING | AI_PROCESSED | SPECIALIST_REVIEW | COMPLETED | ESCALATED
urgency_level	ENUM	NOT NULL	ROUTINE | URGENT | CRITICAL
symptoms	TEXT	NOT NULL	Detailed symptom narrative
vitals_json	JSONB	NULLABLE	Structured vital signs data
lab_results_json	JSONB	NULLABLE	Structured laboratory values
ai_report_id	UUID	FK → DiagnosisRec, NULLABLE	Associated AI diagnostic report
submitted_at	TIMESTAMP WITH TIME	NOT NULL	Submission timestamp (UTC)

Attribute	Data Type	Constraint	Description
	ZONE		
completed_at	TIMESTAMP WITH TIME ZONE	NULLABLE	Resolution timestamp (UTC)

Table 4: DIAGNOSIS_RECOMMENDATION

Attribute	Data Type	Constraint	Description
recommendation_id	UUID	PK, NOT NULL	Unique recommendation identifier
consultation_id	UUID	FK → Consultation, NOT NULL	Parent consultation
diagnosis_name	VARCHAR(255)	NOT NULL	Clinical diagnosis label
icd10_code	VARCHAR(10)	NOT NULL	ICD-10-CM code
confidence_score	DECIMAL(5,2)	NOT NULL	AI probability percentage [0,100]
rank_position	INTEGER	NOT NULL	Position in ranked DDx list
supporting_evidence	TEXT	NOT NULL	AI clinical reasoning narrative
recommended_tests	TEXT	NULLABLE	Suggested follow-up investigations
treatment_suggestion	TEXT	NULLABLE	Initial management pathway
model_version	VARCHAR(50)	NOT NULL	AI model version used
generated_at	TIMESTAMP WITH TIME ZONE	NOT NULL	Generation timestamp (UTC)

Table 5: SPECIALIST_REVIEW

Attribute	Data Type	Constraint	Description
review_id	UUID	PK, NOT NULL	Unique review record identifier
consultation_id	UUID	FK → Consultation, NOT NULL	Associated consultation
specialist_id	UUID	FK → User.user_id, NOT NULL	Reviewing specialist
action	ENUM	NOT NULL	APPROVED | MODIFIED | OVERRIDDEN | ESCALATED
specialist_diagnosis	VARCHAR(255)	NULLABLE	Specialist's diagnosis if overriding AI
annotations	TEXT	NULLABLE	Specialist clinical commentary
revised_treatment	TEXT	NULLABLE	Modified treatment plan

Attribute	Data Type	Constraint	Description
is_critical	BOOLEAN	NOT NULL, DEFAULT FALSE	Critical case flag
reviewed_at	TIMESTAMP WITH TIME ZONE	NOT NULL	Review completion timestamp (UTC)

Table 6: MEDICAL_IMAGE

Attribute	Data Type	Constraint	Description
image_id	UUID	PK, NOT NULL	Unique image record identifier
consultation_id	UUID	FK → Consultation, NOT NULL	Parent consultation
image_type	ENUM	NOT NULL	XRAY | CT_SCAN | MRI | ULTRASOUND | OTHER
file_format	VARCHAR(10)	NOT NULL	DICOM | JPEG | PNG
storage_path	VARCHAR(500)	NOT NULL	Encrypted cloud storage object key
ai_findings	TEXT	NULLABLE	AI imaging analysis output narrative
imaging_confidence	DECIMAL(5,2)	NULLABLE	AI imaging analysis confidence score
uploaded_at	TIMESTAMP WITH TIME ZONE	NOT NULL	Upload timestamp (UTC)

Table 7: AUDIT_LOG

Attribute	Data Type	Constraint	Description
log_id	UUID	PK, NOT NULL	Unique log entry identifier
actor_id	UUID	FK → User.user_id, NOT NULL	User who performed the action
action_type	VARCHAR(100)	NOT NULL	Action label: LOGIN | SUBMIT | VIEW | MODIFY | DELETE
entity_type	VARCHAR(100)	NOT NULL	Affected entity class name
entity_id	UUID	NULLABLE	Identifier of affected entity
ip_address	VARCHAR(45)	NOT NULL	Requestor IP (supports IPv6)
event_timestamp	TIMESTAMP WITH TIME ZONE	NOT NULL	Event time in UTC (immutable)
metadata_json	JSONB	NULLABLE	Additional structured event context

7.4 User Characteristics

The AI-MSOS must accommodate users with markedly heterogeneous technical backgrounds, clinical roles, and infrastructure environments.

- General Practitioners operate across a spectrum of digital proficiency levels and infrastructure conditions. Many work from Android smartphones on 2G or 3G networks in rural facilities with intermittent power supply. The platform must therefore prioritise low-bandwidth performance, intuitive form design, and graceful degradation under poor connectivity. First-time users should complete a consultation submission without external training.
- Specialist Consultants are senior clinicians accustomed to professional software environments, typically with reliable internet connectivity and higher-specification computing equipment. They engage with the platform in structured review sessions and require high-information-density interfaces that enable rapid assessment of AI recommendations and case data.
- System Administrators are technically proficient personnel responsible for user provisioning, platform governance, and regulatory reporting. They require full access to operational dashboards, audit trail query interfaces, and model management controls.

7.5 Assumptions and Dependencies

7.5.1 Assumptions

- All GP users have access to a mobile device capable of running the PWA, and intermittent 2G/3G connectivity is available during case submission; offline queuing covers periods of complete disconnection
- Medical imaging files uploaded by GPs have been acquired on calibrated, properly maintained imaging equipment and meet minimum diagnostic adequacy standards
- Clinical training datasets used in AI model development are demographically and epidemiologically representative of the disease burden and patient populations in target deployment regions
- Institutional administrators in each deploying organisation will assume responsibility for staff onboarding, local regulatory compliance validation, and ongoing user support
- Specialist consultants will honour the service level agreement commitments specified in their onboarding agreement, responding within four hours for routine cases and 30 minutes for critical cases

7.5.2 Dependencies

- Third-party AI and NLP model providers (Hugging Face, Google DeepMind Med-PaLM API) must maintain API availability, version stability, and output consistency; any deprecation requires notification at minimum 90 days in advance
- Cloud healthcare data platforms (AWS HealthLake or GCP Healthcare API) must retain HIPAA Business Associate Agreement coverage and GDPR adequacy decisions for all target deployment jurisdictions

- Partner EHR systems must expose or be capable of exposing FHIR R4-compliant APIs for patient data exchange; where legacy systems require middleware adaptation, institutional IT teams bear responsibility for the adaptation layer
- Mobile platform distribution channels (Google Play Store and Apple App Store) must continue to accept and approve applications classified under their respective health and medical application policies
- Access to curated, validated, de-identified clinical datasets from partner institutions is required for AI model retraining cycles; dataset sharing agreements must be in place prior to Phase 2 model development activities
- Regulatory classification of the AI-MSOS as a clinical decision support tool rather than an autonomous diagnostic device by applicable national health authorities is assumed; if reclassification occurs, additional clinical evidence requirements may apply

DOCUMENT REVISION HISTORY

Version	Date	Author	Change Summary
1.0	April 2026	Development Team	Initial SRS — All 7 sections, 6 system diagrams, 7 database tables